

AMPL Training Series · Part 1 of 2



AMPL

The Modern AMPL Stack

From the legacy IDE to Python-first workflows for teaching, research, and the road to industry.

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THE HOUR AHEAD

Today's Agenda

- 01 **Why the stack moved**

- 02 **The AMPL Python API**

- 03 **Data & solver integration**

- 04 **Toolkit + live demo**

- 05 **Academic licensing**

Quick check — where are you?

AUDIENCE POLL · ZOOM

- A** Legacy IDE only
- B** Exploring Python tooling
- C** Already teaching with notebooks
- D** Mixed across courses

Drop a reaction or use the Zoom poll.

SECTION 01

Why the stack moved

SECTION 01 · WHY

What changed around AMPL

Python everywhere

The lingua franca of computational coursework — across OR, data science, and engineering.

Jupyter in the classroom

Narrative, code, and results in one artifact — the default form factor for assignments.

Cloud-first students

Chromebooks, no admin rights, the expectation that things just run in any browser tab.

Your AMPL hasn't changed. The world around it has.

Your .mod files still work

LEGACY AMPL · CLASSIC IDE

```
● ● ● PRODUCTION.MOD

var x {P} >= 0;

maximize Profit:
    sum {p in P} margin[p] * x[p];

subj to demand {p in P}:
    x[p] <= cap[p];
```

MODERN STACK · JUPYTER CELL

```
● ● ● IN [3]: NOTEBOOK.IPYNB

%%ampl_eval
var x {P} >= 0;

maximize Profit:
    sum {p in P} margin[p] * x[p];

subj to demand {p in P}:
    x[p] <= cap[p];
```

The AMPL language is identical. What's new is everything wrapping it — Python, notebooks, package management, cloud.

A better toolbox

Editors

VS Code · amplpy · Jupyter

Use as much or as little of the modern stack as fits your course.

Cloud

Google Colab · MO-Book

Zero install. Runs in any browser tab.

Distribution

pip · Portal · amplbot

One command anywhere — Mac, Windows, Linux, Colab.

SECTION 02

What the modern toolbox looks like

SECTION 02 • THE PYTHON API

What amplpy does

AMPL ↔ PYTHON • LOAD DATA

```
LOAD_DATA.PY

from amplpy import AMPL
import pandas as pd

ampl = AMPL()
ampl.read('production.mod')

df = pd.read_csv('demand.csv')
ampl.set_data(df, 'P')
ampl.solve(solver='highs')
```

AMPL ↔ PYTHON • READ RESULTS

```
READ_RESULTS.PY

x = ampl.get_variable('x')
results = x.get_values().to_pandas()

results.plot.bar()
plt.title('Production plan')

print(ampl.get_objective('Profit').value())
```

Pandas in. Pandas out. The model is the only thing in AMPL syntax.

SECTION 02 · PYTHON API

Why this matters for teaching and research

Version control

Notebooks and .mod files in Git — full diff history on student assignments.

Parameter sweeps

Solve hundreds of variants from a Python loop. No hand-edited .dat files.

Sharable artifacts

One .ipynb runs on a laptop, a Google Colab tab, or a cloud instance.

Stop writing **.dat** files

⚠ CLASSICAL · PLANTS.DAT

```
● ● ●                                ⚠ PLANTS.DAT

set Plants :=
    Detroit  Pittsburgh  Atlanta;

param capacity :=
    Detroit      900
    Pittsburgh    1200
    Atlanta      800;
```

MODERN · PANDAS DATAFRAME

```
● ● ●                                IN [4]: NOTEBOOK.IPYNB

import pandas as pd
# change to amply
df = pd.read_csv("plants.csv")
ampl.set_data(df, "Plants")

# now drives the same model
```

Reach for pandas when your data lives in CSVs, databases, or APIs — not in **.dat** files.

SECTION 03

How AMPL plugs into Python

Installing solvers is just **pip** now

```
~/COURSE — TERMINAL  
  
$ python -m pip install amplpy --upgrade  
$ python -m amplpy.modules install highs gurobi xpress cplex  
$ python -m amplpy.modules activate <your-license-uuid>
```

No PATH wrangling

No separate downloads

Same command on Mac, Windows, Linux, Colab

Three commands. Same on Mac, Windows, Linux, and Colab.

Swap solvers without changing the model

```
IN [7]: NOTEBOOK.IPYNB

ampl.solve(solver=" highs")      # open source
ampl.solve(solver="gurobi")     # commercial
ampl.solve(solver="cplex")      # commercial
```

AMPL's MP driver also gives better behavior on logical and conditional constraints — your students' first instinct usually compiles.

SECTION 04

What AMPL ships in the toolkit

Tools in your students' hands

VS Code + AMPL extension

Syntax highlighting, debugging, integrated terminal, Git — the editor your students already know.

Jupyter Notebook + Colab

Narrative + code + output in one artifact. Zero-install via Colab kills the 'doesn't work on my machine' problem.

MO-Book + amplbot

Free, Colab-runnable optimization textbook + AI assistant trained on AMPL syntax.

LIVE DEMO

MO-Book Ch.1 — Production planning

<https://ampl.com/mo-book/notebooks/01/production-planning-advanced.html>



SECTION 04 • TOOLKIT

What just happened

- ✓ Same AMPL syntax — var, maximize, subj to
- ✓ Model is solver-agnostic
- ✓ Zero local install — your students did it from a browser

Default community license got us here. Let's talk about what's next.

SECTION 05

What this means for your team

AMPL travels with your students

<https://ampl.com/academia/>

A4A Academics

- Anyone with academic email
- 10 commercial + open source solvers
- No time limit, 12 months renewal
- Self-serve in Portal
- Free, individual use

A4C Courses

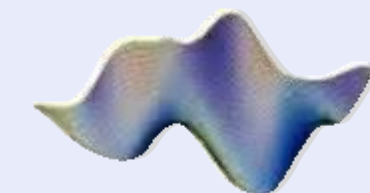
- Instructor-requested, distributable
- All commercial solvers, no fees
- Up to 8 months
- Shareable activation link
- Covers the whole class

<https://portal.ampl.com/user/ampl/request/ampl-for-courses/new>

ACS

- Recent grads (≤ 24 months)
- All commercial solvers
- 12 months · commercial OK
- Bridges classroom → industry

<https://portal.ampl.com/user/ampl/request/ampl-career-starter/new>



SECTION 05 · LICENSING

Where to start

- A** Portal home — sign in or create an account with your academic email
- B** A4A self-serve form — instant individual academic license
- C** A4C course request — instructor-only; returns a shareable activation link

portal.ampl.com/account/ampl/

COMING UP · PART 02

Part 2 — Hands-on

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PRE-WORK CHECKLIST

- Laptop with Python 3.10+ (preferably)
- Visual Studio Code installed (recommended)
- A .mod file from your course
- Academic license already activated

WHAT YOU'LL DO

- Install amplpy + VS Code extension
- Run diet.mod as a baseline
- Switch HiGHS → Gurobi via your A4A